IT-SIMPLERA INSTITUTE -CYBER SECURITY ANALYST INTERNSHIP PROGRAM

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02th July, 2026

Task 1: Research Five Real-World Cyberattacks

Attack Name	Year	Organization	Attack Type	Description	Impact
WannaCry ransomware attack	2017	National Health Service and others	Ransomware	Exploited a Windows vulnerability and encrypted files on infected systems.	Over 200,000 systems affected in 150+ countries; significant operational disruption.
Equifax data breach	2017	Equifax	Data Breach	Attackers exploited an unpatched web application vulnerability.	Personal information of approximately 147 million people exposed.
SolarWinds supply chain attack	2020	SolarWinds	Supply Chain Attack	Attackers inserted malicious code into software updates.	Multiple government agencies and private companies compromised.
Colonial Pipeline ransomware attack	2021	Colonial Pipeline	Ransomware	Attackers gained access using compromised credentials and deployed ransomware.	Fuel distribution disruptions across the United States.
Target data breach	2013	Target Corporation	Data Breach	Attackers compromised a third-party vendor and gained access to payment systems.	Millions of customer payment card records stolen.

Task 2: CIA Triad Analysis

1. WannaCry Ransomware Attack

CIA Component Violated: Availability

Explanation:

The ransomware encrypted files and prevented users from accessing their systems and data, making services unavailable.

2. Equifax Data Breach

CIA Component Violated: Confidentiality

Explanation:

Sensitive personal information was accessed and stolen by unauthorized individuals.

3. SolarWinds Supply Chain Attack

CIA Components Violated: Integrity and Confidentiality

Explanation:

Malicious code was inserted into legitimate software updates, compromising software integrity and allowing attackers access to sensitive information.

4. Colonial Pipeline Attack

CIA Component Violated: Availability

Explanation:

Pipeline operations were disrupted, preventing normal service delivery.

5. Target Data Breach

CIA Component Violated: Confidentiality

Explanation:

Customer payment card data and personal information were stolen by attackers.

Task 3: Research Three Cybersecurity Tools

1. Wireshark

Purpose: Network traffic analysis

Function: Captures and analyzes network packets in real time.

Use Case: Investigating suspicious network activity and troubleshooting network issues.

2. Nmap

Purpose: Network discovery and security auditing.

Function: Identifies hosts, services, operating systems, and open ports.

Use Case: Asset discovery and vulnerability assessment.

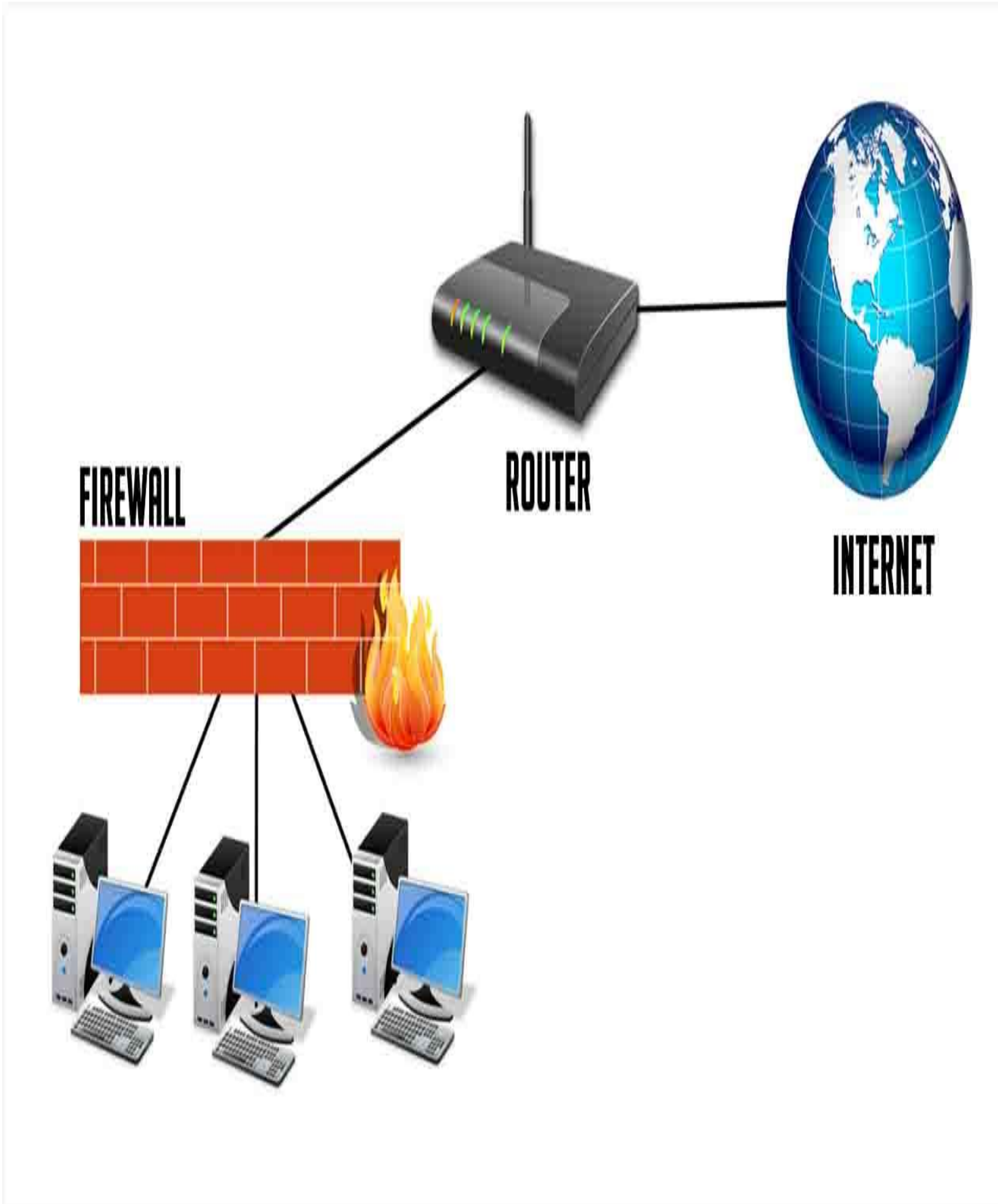
3. Splunk

Purpose: Security monitoring and log management.

Function: Collects, indexes, and analyzes security logs from multiple sources.

Use Case: Threat detection, incident investigation, and compliance reporting.

Task 4: Secure Home Network Diagram



Task 5: What Does a Cybersecurity Analyst Do Every Day?

A Cybersecurity Analyst is responsible for protecting an organization's systems, networks, and data from cyber threats. The role involves continuous monitoring of security alerts, investigating suspicious activities, identifying vulnerabilities, and responding to security incidents.

A typical day begins with reviewing alerts generated by SIEM platforms, firewalls, intrusion detection systems, and endpoint security tools. Analysts determine whether alerts represent real threats or false positives and prioritize incidents based on severity.

Cybersecurity analysts investigate security incidents by examining logs, network traffic, user activity, and endpoint events. They gather evidence, determine the root cause of incidents, and recommend containment and remediation actions.

Another important responsibility is vulnerability management. Analysts review vulnerability scan results, identify high-risk weaknesses, and work with system administrators to ensure security patches are applied promptly.

Threat intelligence is also a key part of the job. Analysts stay informed about emerging threats, malware campaigns, vulnerabilities, and attacker techniques. This information helps organizations proactively improve their defenses.

Cybersecurity analysts document their findings, prepare incident reports, and communicate security risks to management and technical teams. Accurate documentation supports compliance requirements and helps improve future incident response efforts.

Overall, cybersecurity analysts play a critical role in maintaining the confidentiality, integrity, and availability of organizational information systems while helping defend against evolving cyber threats.

Challenges Encountered

- Understanding the impact of different cyberattacks.
- Mapping attacks to CIA Triad principles.
- Learning how cybersecurity tools operate.
- Designing a secure network architecture.
- Researching current cybersecurity threats and trends.

Conclusion

This assignment provided practical knowledge of cybersecurity fundamentals, major cyberattacks, the CIA Triad, security tools, secure network design, and the responsibilities of cybersecurity analysts. These concepts form the foundation for future cybersecurity studies and professional development.

References

NIST Cybersecurity Framework

CISA Cybersecurity Resources

Wireshark Documentation

Nmap Documentation

Splunk Documentation